

Lucas Chiang

12203 Linden Linthicum Lane, Clarksville, Maryland 21029 • 410-782-1941 • lucaschiang07@gmail.com

EDUCATION

River Hill High School, Clarksville, Maryland; Class of 2026, Current Weighted/Unweighted GPA: 4.83/3.96

Relevant Coursework: AP Physics C Mechanics, AP Physics C Electricity and Magnetism, AP Calculus AB, AP Calculus BC, AP Chemistry, AP Computer Science Principles, AP Biology (current), Differential Equations (current)

STEM Academy at Advanced Physics Laboratory, 11100 Johns Hopkins Rd, Laurel, Maryland 20723, enrolled since Fall 2022

Relevant Coursework: Critical Thinking 1, Python 2, Python 3, Systems Tool Kit (STK) 1

RESEARCH

Research Intern, Department of Mechanical Engineering, University of Maryland, Baltimore County Summer 2025 - Current

- Created circuitry to power a heart sensor using integrated circuits and wearable thermoelectric generators;
- Patel A, Lombardo J, **Chiang L**, Madabushi V, Kulkarni S, Madan D (2025). "Designing and Constructing a Thermoelectric Powered Self-Sustaining Sensor Platform." Poster presented at the University of MD Summer Undergraduate Research Festival.

Research Intern, Department of Physics, Towson University Summer 2025

- Created thin metal oxide films using Pulsed Laser Deposition, and tested resistance;
- **Chiang L**, Toomer C, Previti M, Pokhrel S, Kolagani R (2025). "Modulation of Electrical Transport and Structural Properties of CaMnO_3 Epitaxial Thin Films." Poster presented to the research team at Towson University.

NASA Data Science Analysis and Scholarly Research Paper Writing Intern, George Mason University Summer 2024

- Conducted independent research analyzing ground-based data from the George Mason Telescope to determine whether an object of interest is an exoplanet;
- **Chiang L**, Plavchan P (2024). "Ground-based Light Curve Follow-up Validation Observations of TESS Object of Interest (TOI) 5868.01." Published in the SCHAR AstroScholars Journal and the George Mason MARS Repository.

LEADERSHIP EXPERIENCES

Design Leader and Outreach Lead, FIRST Tech Challenge (FTC) Team 18996 2020 - Current

- Lead meetings with design team to come up with requirements for subsystem designs for all the robots for the past 5 years;
- Lead outreach activities for the past 3 years: these include heading a team of four members to send emails to more than 100 possible partners and sponsors; organizing three weekly educational events with partners and meet-ups with sponsors to discuss details; teaching STEM principles to underserved kids in Baltimore, partnering with the non-profit Dayspring;
- Team advanced to the Chesapeake FTC Regional Competition; was a Semi-Finalist in 2023 and 2024; received Think Award.

Co-Founder and Vice President, Economics Club, River Hill High School Fall 2023 - Current

- Co-founded club in 10th grade and organized four teams to compete in two divisions for the statewide economics competition; placed second in the Statewide Economics Challenge since the first year of club establishment (Spring 2024, Spring 2025);
- Created multiple curriculums to guide groups of students with different levels of understanding in economics.

Co-Founder and Co-President, Engineering Club, River Hill High School Fall 2024 - Current

- Co-founded club and teach Computer Aided Design using Fusion 360 to club members.

WORK EXPERIENCE

Swim Instructor, Columbia Association 2023 - 2025

Taught children how to swim; monitored progress of children over a 6-week period to create more in-depth lessons.

AWARDS and HONORS

National Merit Commendable, 2025; Honor Roll every quarter in high school

Maryland Technology Honor Society (MTHS) Vice President, 2023 - 2025

Tri-M Music Honor Society Member, since 2023; Junior All-State Band, tenor saxophone, 1st chair, 2022

Received Varsity Letter from Cross Country and Track every year in high school; selected to be Captain in the senior year